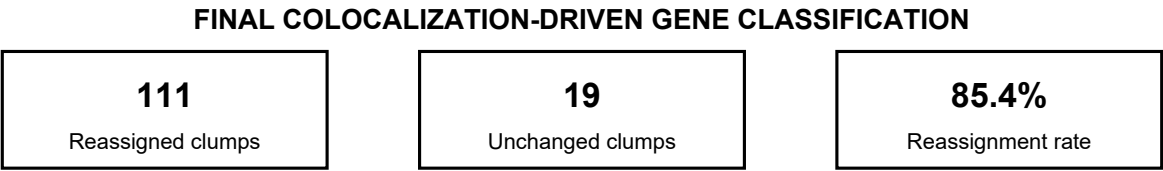


Quality control and analytical summary of the PigmentationAtlas pipeline

Sequential audit of variant harmonization, signal definition, gene annotation, molecular QTL integration and effector-gene prioritization

ANALYTICAL STEP	PIPELINE OUTPUT	QUALITY-CONTROL AUDIT	STATUS
GWAS harmonization	21.8 million harmonized variants	Variant identifiers, alleles, positions and association statistics harmonized	PASSED
Genome-wide signal selection	33,455 genome-wide significant variants	Variants retained at $P < 5 \times 10^{-8}$	PASSED
LD clumping	1,325 independent PLINK clumps	Independent association signals defined using the EUR LD reference panel	PASSED
Genomic locus construction	194 merged genomic loci	Complete locus construction: 0 empty loci; 0 missing lead variants	PASSED
Candidate-gene annotation	2,024 candidate genes	GENCODE v50 gene annotation and biotype assignment	PASSED
Molecular QTL colocalization	9,814 completed colocalization analyses	H4 evidence: 30 strong, 54 moderate and 239 suggestive analyses	PASSED
Effector-gene prioritization	130 H4-supported evaluable clumps	111 reassigned; 19 unchanged (85.4% reassignment)	PASSED



All seven analytical stages completed successfully. Among 130 evaluable H4-supported clumps, 111 (85.4%) were assigned to a gene different from the proximity-based baseline.